Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_replay_solver_evolution.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.15406	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	75.79714	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1387.7622	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	3.0	n/a	209.53802	1	0	0	0.963614	14.28
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	198.07736	1	0	0	0.943062	15.45
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.070755	0.070755	723.65742	2	0	0	0.80531	11.415

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.15406.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 75.79714.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1387.7622.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 209.53802.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 198.07736.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.065064.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.070755.
- Held-out runtime ms: 723.65742.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.003603.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.098678, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.095697, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.059835, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.05446, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.045107, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Aspect	Observations
Direct Synthesis	*phase9_closeout_direct_generate_plus_one_repair* had zero feasibility (0.0) on heldout and high penalized gap (3.0), indicating failure to produce valid solutions.
Budget-Matched Independent Search	*phase9_closeout_budget_matched_no_replay* similarly had 0% feasibility and a large penalized gap (3.0), also failing to find feasible solutions.
Replay-Aware Iterative Search	*phase9_closeout_replay_solver_evolution* showed strong performance with 100% feasibility, a low feasible gap (0.070755), and reasonable runtime (~724 ms). This method is the best heldout condition.
Fixed-Baseline Context	Baselines all had 100% feasibility: - Nearest Neighbor: gap 0.2734, runtime 42 ms - Clarke-Wright Savings: gap 0.0738, runtime 76 ms - Regret Insertion Local Search: gap 0.4664, runtime 1388 ms
Feasibility	Only replay-aware search and fixed baselines achieved 100% feasibility. Direct synthesis and budget-matched no-replay had 0%.
Objective Gap	*Replay solver evolution* (0.0708) slightly better than best baseline Clarke-Wright (0.0738). Other baselines notably worse or infeasible.
Runtime	Replay (~724 ms) slower than Clarke-Wright baseline (~76 ms) but faster than regret insertion (~1388 ms). Nearest neighbor fastest (~42 ms) but worse gap than replay.

Aspect	Observations
Robustness	Replay solver evolution had low robustness dispersion (0.0036), better than Clarke-Wright (0.0078) and much better than regret insertion (0.0869), indicating consistent results across structure families.
Learned Condition vs. Baselines	Replay solver evolution clearly outperformed all fixed baselines on feasible gap and robustness while maintaining full feasibility. Direct synthesis and budget-matched no-replay did not produce feasible solutions.
Replay vs. Budget-Matched No-Replay	Replay solver evolution strongly outperformed budget-matched no-replay control, which had zero feasibility and maximal penalized gap (3.0).

Conclusion:

- **Replay-aware iterative search (phase9_closeout_replay_solver_evolution) was the only learned method to clearly beat fixed baselines**, delivering better feasible gaps, full feasibility, and strong robustness despite higher runtime.
- **Direct synthesis and budget-matched independent search failed to produce feasible solutions.**
- **Replay strategy significantly outperformed the budget-matched no-replay control, confirming the value of replay in solving the CVRP here.**